

QQQ volume during the MNQ bot trades

How the check was built, what it showed, and how to run it again. September 30, 2026.

This note pairs each counted MNQ bot round trip with the QQQ share volume that was printing on the 5-minute candles while that trade was open. The question was simple: do the losing trades sit on quieter QQQ candles than the winning trades?

On the trades that actually overlap the QQQ regular session, losers did sit on quieter candles. The link is real and it is weak. Heavier candles won more often. They did not make more money. This is a research check on a dry-run log. It is not a live filter.

The two files

Both files go with this PDF. The agent should use these files and no others. Times in the trade log are already Central time. Do not shift them.

1. Trade log — MNQ_Bot_PnL.xlsx, sheet named Trades. One row is one round trip. Use only rows where the column Count in totals? is Y. That set is 149 trips, from August 23 through September 30, 2026. Rows marked N are the September 3 replay burst. Every fill in that burst is stamped 29,623.25. They are not real prices and they stay out. Net P&L; is already in dollars. The sheet uses \$2 per MNQ point and \$0 fees, so net equals gross. A winner is net above zero. A loser is net below zero. There are no flat trips in the counted set.

2. QQQ candles — the TradingView export of NASDAQ:QQQ, 5-minute, regular hours. Columns are time, open, high, low, close, Volume. time is a UNIX timestamp for the **start** of the candle. Volume is QQQ shares traded in that candle only, not a running total and not MNQ contracts. This export has 3,310 candles, from July 31, 2026 at 12:10 CT through September 30, 2026 at 14:55 CT. A full day is 78 candles, 8:30 a.m. to 3:00 p.m. Central. There is no premarket and no overnight in this file.

How a trade was paired with volume

For each counted trip, take every 5-minute QQQ candle that overlaps the open trade.

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A candle overlaps when
  candle_start < exit time
  and candle_start + 5 minutes > entry time
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Average volume = the mean of Volume on those candles
Usual volume    = for that same clock minute, the median Volume
                  of every day in the QQQ file
                  (10:30 is compared with other 10:30s, not with the open)
Baseline        = the mean of those usual volumes, for the candles in the trade
Ratio           = average volume / baseline
                  1.00 = normal for that time of day
                  1.50 = 50 percent heavier than usual for those minutes
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A trade is in the main table only if those candles cover at least 80 percent of the minutes it was open. Coverage is the overlapping candles times 5 minutes, divided by how long the trade lasted, and it cannot exceed 100 percent. A 3-minute trade that sits inside one candle still counts. The whole candle is used in the average, because that candle was printing while the trade was on.

Trades with no overlapping candle are outside this file. They are overnight, premarket, or a Sunday futures session. They are reported separately and they are not mixed into the win-rate table. Trades that only partly overlap the regular session, under 80 percent, are also kept out of the main table.

What was left in the comparison

Group	Trades	Winners	Losers	Net P&L;
Counted trips (Count = Y)	149	59	90	−\$5,867
No QQQ candle at all	78	30	48	−\$2,746
Only partly inside the session	10	5	5	+\$795
Main set, 80 percent or more	61	24	37	−\$3,917

The whole log is −\$5,867. The 61 trades we can actually score are −\$3,917. The 78 trades with no QQQ candle are −\$2,746. This volume check cannot explain those 78, because the export has no candles for them.

The result

Main set only. Average QQQ shares per 5-minute candle while the trade was open.

	Trades	Median 5-min volume	Mean 5-min volume	Median ratio	Net P&L;
Winners	24	242,606	295,405	1.13	+\$14,334
Losers	37	197,945	250,055	0.99	−\$18,250

Losers printed on quieter candles: about 197,945 shares in a 5-minute bar, against about 242,606 for winners. A typical 5-minute bar in this whole export is about 199,000 shares, and a normal day is about 20 million shares.

Pearson correlation of average volume with win or loss is **0.10**. Correlation with dollars is **0.09**. After the clock adjustment, correlation of the ratio with win or loss is **0.09**, and with dollars **0.03**. A useful standalone filter would be much stronger than this. Around 0.10 is a tilt, not a switch.

Same 61 trades, split into four equal volume groups

The cuts are the 25th, 50th, and 75th values after the 61 averages are sorted. With 61 numbers, those positions are index 15, 30, and 45, counting from zero. The cut values are 156,613, 206,955, and 277,092 shares.

Volume while the trade was open	Trades	Wins	Losses	Win rate	Net P&L;
Quietest · 156,613 or less	16	4	12	25%	−\$2,547
Low-mid · up to 206,955	15	5	10	33%	−\$393
High-mid · up to 277,092	15	7	8	47%	−\$1,366
Heaviest · above 277,092	15	8	7	53%	+\$390

Same trades, but each candle is compared with the usual volume at that clock time

This takes the open out of the comparison. A heavy open candle is normal at 8:30. It only counts as heavy here if it is heavy versus other days at that same minute. Ratio cuts are 0.89, 1.03, and 1.37.

Versus the usual volume at that time	Trades	Wins	Losses	Win rate	Net P&L;
Quiet vs that clock · 0.89 or less	16	4	12	25%	−\$2,214
A bit light · up to 1.03	15	5	10	33%	−\$2,828
A bit heavy · up to 1.37	15	7	8	47%	+\$1,568
Heavy vs that clock · above 1.37	15	8	7	53%	−\$442

Reading both tables the same way: the quiet group won about 1 trade in 4. The heavy group won about 1 in 2. The heavy group still did not pay. On raw volume it was about flat (+\$390). Versus the clock, the heaviest group lost \$442. Quiet tape is where the losses bunch up. Heavy tape is not, by itself, a reason to take the trade.

Two trades, candle by candle

These two are in the main set so a second person can add the volumes by hand and match the average.

A winner

Trade #211. Long, 7 contracts. September 24, 11:15 to 11:20 CT. Net +\$1,852. 2 candles. Average 839,984. Usual for those minutes 170,431. Ratio 4.93.

candle start CT	QQQ shares	usual shares at this minute
09-24 11:15 volume	1,185,508	usual at this minute 158,911
09-24 11:20 volume	494,459	usual at this minute 181,952
mean of the volume column = 839,983.5		

A loser

Trade #225. Short, 7 contracts. September 30, 09:35 to 09:50 CT. Net -\$808. 4 candles. Average 178,976. Usual for those minutes 258,023. Ratio 0.69.

candle start CT	QQQ shares	usual shares at this minute
09-30 09:35 volume	180,447	usual at this minute 280,932
09-30 09:40 volume	223,973	usual at this minute 254,716
09-30 09:45 volume	127,904	usual at this minute 255,150
09-30 09:50 volume	183,579	usual at this minute 241,294
mean of the volume column = 178,975.8		

The same heavy print can be either. September 16 at 1:50 p.m. was about 5 times the normal volume for that minute and lost \$1,392. September 24 at 11:15 a.m. was also a heavy two-candle burst and made \$1,852. One large candle is not a signal by itself.

What this does not say

- It does not say the bot should stop trading quiet QQQ. The sample is 61 dry-run trips over one month. A 25 percent win rate against a 53 percent win rate is a tilt. It is not a tested rule.
- It does not explain the trades outside regular QQQ hours. Those 78 trips are a separate -\$2,746 and this file cannot see them.
- It does not say QQQ share volume is the same thing as MNQ contracts or as the bot's order-flow delta. This is the cash-index ETF, 5-minute shares only.
- It does not say last week was a low-volume regime and this week was a high-volume regime. On this export, the week of September 14 was about 111 million QQQ shares and the week of September 21 was about 108 million. Those two weeks are the same size.
- Dollars are rounded to the nearest dollar for reading. The workbook has floating-point noise in a few net figures. Fees are zero in the sheet, so nothing here is after commission.

Instructions for the agent

Paste the block below to the agent that will rerun this. Give it this PDF plus the two files. The agent's job is to reproduce the tables, not to invent a new trading rule.

TASK

Reproduce the QQQ 5-minute volume check on the MNQ bot round trips.

Do not change the rules below. If a file column is missing, stop and say so.

FILES

1. MNQ_Bot_PnL.xlsx

Sheet: Trades

Header row contains: #, Session, Contract, Side, Contracts, Entry time (CT), Avg entry, Exit time (CT), Avg exit, Points, Gross P&L, Fees, Net P&L, Cumulative net, Exit reason, Count in totals?

Data starts on the next row.

Keep a row only when Count in totals? is exactly Y.

Entry time and Exit time are already America/Chicago.

Do not convert them and do not apply a timezone offset.

Winner: Net P&L > 0. Loser: Net P&L < 0.

2. QQQ 5-minute export. CSV or xlsx, same columns:

time, open, high, low, close, Volume

time is UNIX seconds of the candle START.

Convert time with zone America/Chicago, then drop tzinfo so it can be compared with the naive Excel datetimes.

Volume is QQQ shares in that candle. Do not cumulate it.

PAIRING

A candle overlaps a trade when

candle_start < exit

AND candle_start + 5 minutes > entry

Average volume = mean of Volume on overlapping candles.

For each clock (hour, minute), usual volume = median Volume

of every candle in the QQQ file with that same hour and minute.

Baseline = mean of those usual volumes for the overlapping candles.

Ratio = average volume / baseline.

Duration minutes = (exit - entry) in minutes.

Coverage = min(bar_count * 5, duration) / duration.

Main set = coverage >= 0.80 and at least one overlapping candle.

Do not put the other trades in the win-rate table.

QUARTILES

Sort the main-set average volumes.

Cut = value at index round((n - 1) * p), p = 0.25, 0.50, 0.75.

Q1: avg <= cut25

Q2: cut25 < avg <= cut50

Q3: cut50 < avg <= cut75

Q4: avg > cut75

Do the same sort-and-index cuts on the ratio.

REPORT

1. Counts: all Y rows, no candle, partial coverage, main set.

Winners, losers, and net P&L for each group.

2. Main set only: winners vs losers, median and mean average volume, median ratio, net P&L.

3. Pearson correlation of average volume with a 1/0 win flag, and with net dollars. Same two correlations for the ratio.

4. The four volume groups and the four ratio groups:

trades, wins, losses, win rate, net P&L.

5. One winning trade and one losing trade from the main set, listing each overlapping candle's start time and Volume, so the average can be checked by hand.

DO NOT

- Do not score Count = N.

- Do not use daily volume, or only the entry candle.

- Do not drop a candle because the trade used part of it.

- Do not call this a live rule or change the bot.

- If your quartile cuts differ from 156,613 / 206,955 / 277,092 shares, or your main set is not 61 trades (24 wins, 37 losses),

stop and show the first five paired rows before changing anything.

Expected main-set check figures, so a rerun can be trusted before anyone reads the prose: 61 trades, 24 winners, 37 losers, winner median volume 242,606, loser median volume 197,945, correlation of average volume with win/loss 0.103.